

IMCA 221
Programming for Artists
Fall 2025

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**Class is on Zoom even in the classroom
for sharing, find the details on Moodle**

<https://moodle.concordia.ca/>

Download the slides!

Inspo

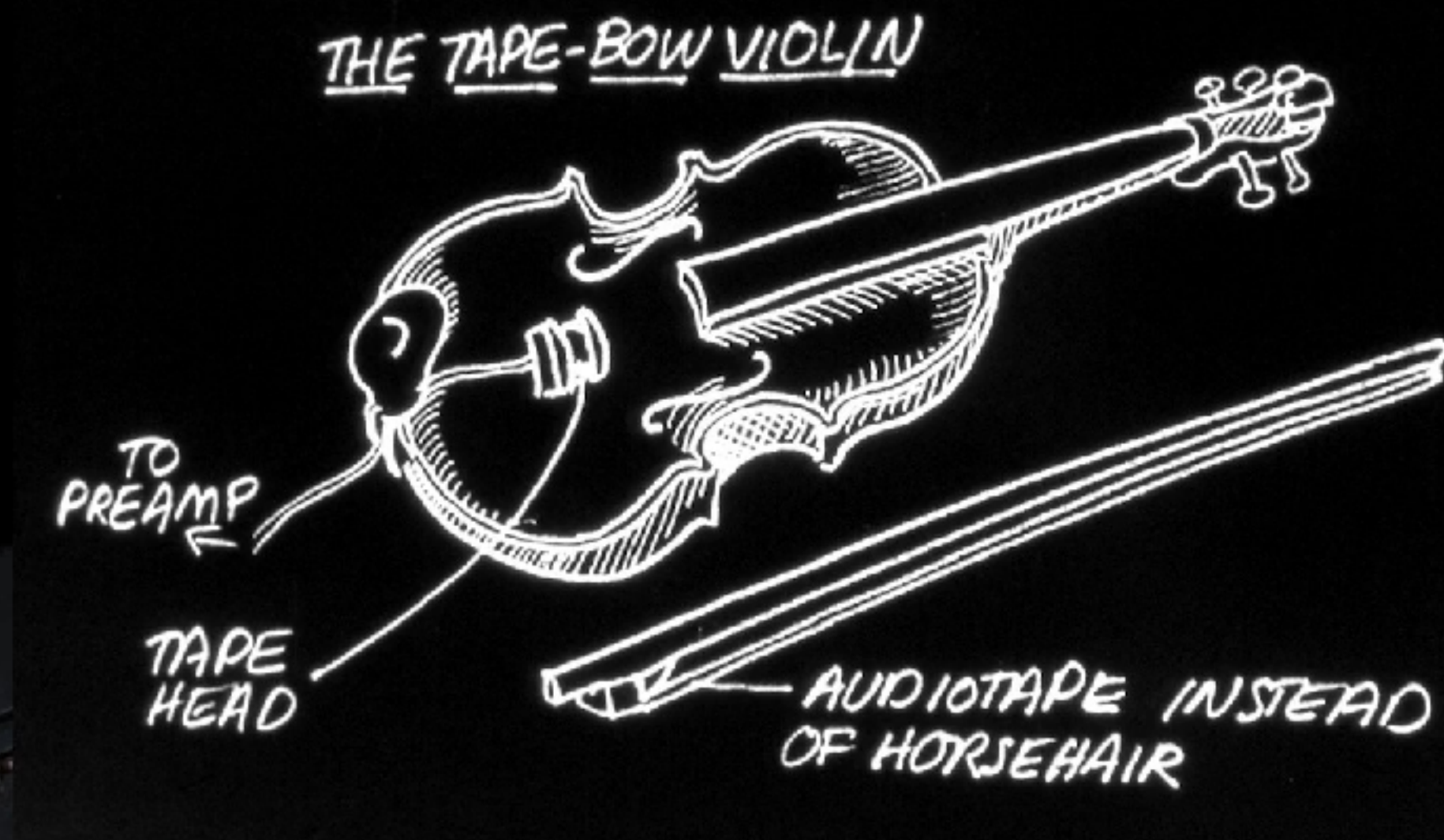


Imagon Heap Gloves

[https://
www.dezeen.com/
2014/03/20/imogen-
heap-funding-drive-
for-gloves-that-turn-
gestures-into-music/](https://www.dezeen.com/2014/03/20/imogen-heap-funding-drive-for-gloves-that-turn-gestures-into-music/)



Victoria Shen
[https://
evicshen.com/
misc](https://evicshen.com/misc)



Laurie Anderson
<https://cdm.link/2021/09/laurie-anderson-in-video-on-the-stories-behind-her-custom-built-instruments/>



Janet Cardiff
[https://
www.youtube.co
m/watch?
v=38ORiaia9r8](https://www.youtube.com/watch?v=38ORiaia9r8)

Resources:

Max Documentation: <https://docs.cycling74.com/max8>

Max Cookbook <https://music.arts.uci.edu/dobrian/maxcookbook/>

Andrew Robinson Video Tutorials <https://www.youtube.com/@AndrewRobinson26>



Max Thesaurus

[https://docs.cycling74.com/max8/
vignettes/thesaurus](https://docs.cycling74.com/max8/vignettes/thesaurus)

Find the name for objects that might be difficult to find otherwise!

Keyboard Shortcuts

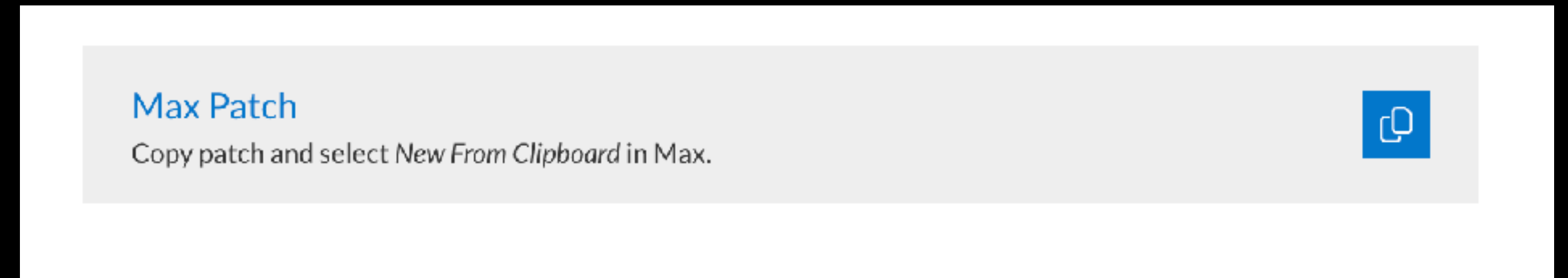
<https://docs.cycling74.com/max8/vignettes/keycommands>

- a: attrui.
- b: button.
- c: comment.
- f: floating point number box.
- h: briefly highlights a small area around the cursor.
- H: A capital letter "H" briefly highlights a larger area around the cursor.
- i: integer number box.
- j: object box containing "jit." for creating Jitter objects.
- l: object box containing "live." for creating Live objects.
- m: message.
- n: new blank object with the cursor active. Typing the name of any object and pressing enter or clicking outside of the object box will transform it into that object.
- r: bring up a list of the most recently created objects, including any arguments and attributes typed in. Selecting an element from the list creates an object with the corresponding text and with the cursor active at the far right. Hitting a carriage return or clicking away from the object instantiates the object.
- p: create an object box containing the message newobj @presentation 1 @text and a cursor. When you type the name of an object (e.g. dial) and hit a carriage return, the object will transform itself into a copy of the object whose name you type in, and the object will be automatically added to the Presentation Layer.
- s: slider.
- t: toggle.
- x: shows a menu describing the key commands, including those added by external packages.
- z: zooms the patcher in around the cursor.
- Z: A capital letter "Z" zooms the patcher out around the cursor.

You can copy Max code and paste it directly into your patch as text.

CMD + V (paste)

```
{
  "boxes" : [ {
    "box" : {
      "maxclass" : "gain~",
      "patching_rect" :
[ 385.792366743087769,
357.923513054847717, 22.0, 140.0 ],
      "outlettype" : [ "signal", "" ],
      "multichannelvariant" : 0,
      "id" : "obj-114",
      "parameter_enable" : 0,
      "numinlets" : 1,
      "numoutlets" : 0
    }
  ]
}
```



Audio Experiment Due next week (midnight, after class)

- Explore audio tools we worked on in class
- Practice and explore tutorials online
- Create an experiment, which could be the beginning of a future project
- Your experiment should have a concept, even though it is not fully developed.

To hand in your project, use the project template on Moodle
saved file (File > Save as Project) .maxpat

- A PDF that contains project documentation. See Project_Documentation_template
- A clear, strong image of your project.
- A screen capture of your max patch
- A link to video or audio recording of your project working (can be a video or a link to a private video on YouTube, Vimeo or Google)
- A 50-100 word explanation of your project inside your maxpatch Be sure to name files properly (no untitled-1.zip) All files are expected to be cleaned up and arranged in a reasonable, legible way. Videos should be clear, well light and show your project working.

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Videos should be clear, well light and show your project working.

My Project Title

Lee Wilkins



Main Image:

Include a clear, strong image of your work. If the work is just audio, include an image of whatever context you imagine it in.

Artist Statement:

Write an artist statement that describes your work. Describe the work as though it is on a gallery wall, what would you read? 50-100 words. Remember, even a small project should have some kind of theme or concept. What do you want people to experience while viewing your work?

A good place to start is:

[Project Name] is an exploration of [concept]. By using [something, audio, video, samples from something?] viewers are able to experience [what do you want them to feel?].

Project Template on
Moodle has
everything you need,
just replace the text
and images with your
own.

Audio Experiment Due next week (midnight, after class)

25% Functionality / technical

25% creativity / concept

25% execution and quality

25% documentation

My Project Title

Lee Wilkins



Main Image:

Include a clear, strong image of your work. If the work is just audio, include an image of whatever context you imagine it in.

Artist Statement:

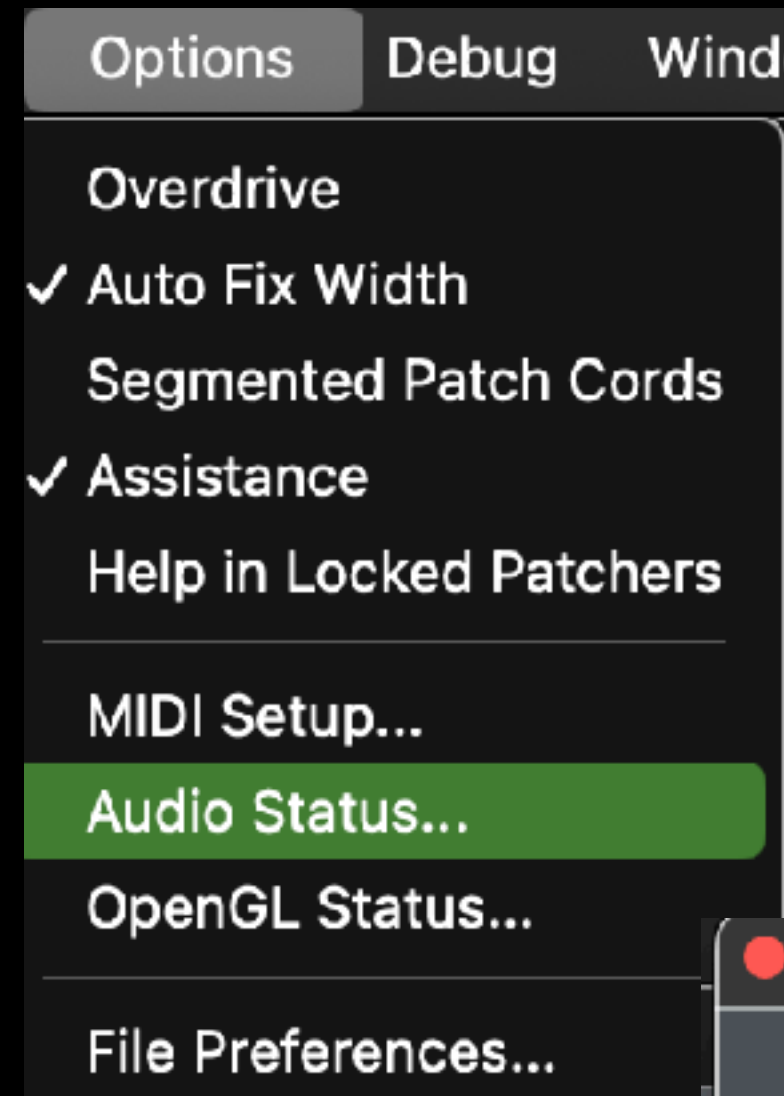
Write an artist statement that describes your work. Describe the work as though it is on a gallery wall, what would you read? 50-100 words. Remember, even a small project should have some kind of theme or concept. What do you want people to experience while viewing your work?

A good place to start is:

[Project Name] is an exploration of [concept]. By using [something, audio, video, samples from something?] viewers are able to experience [what do you want them to feel?].

Project Template on Moodle has everything you need, just replace the text and images with your own.

Audio Status shows where sound is being output. Make sure the sources are correct. You may need to restart it (press the power button)



Ezdac~ is an object
for playing audio

~ means you are
working with an audio
object

Ezdac has two inputs,
left and right channel.



On the sidebar click the music note.

Find a tune you like, drag and drop it into your patch.

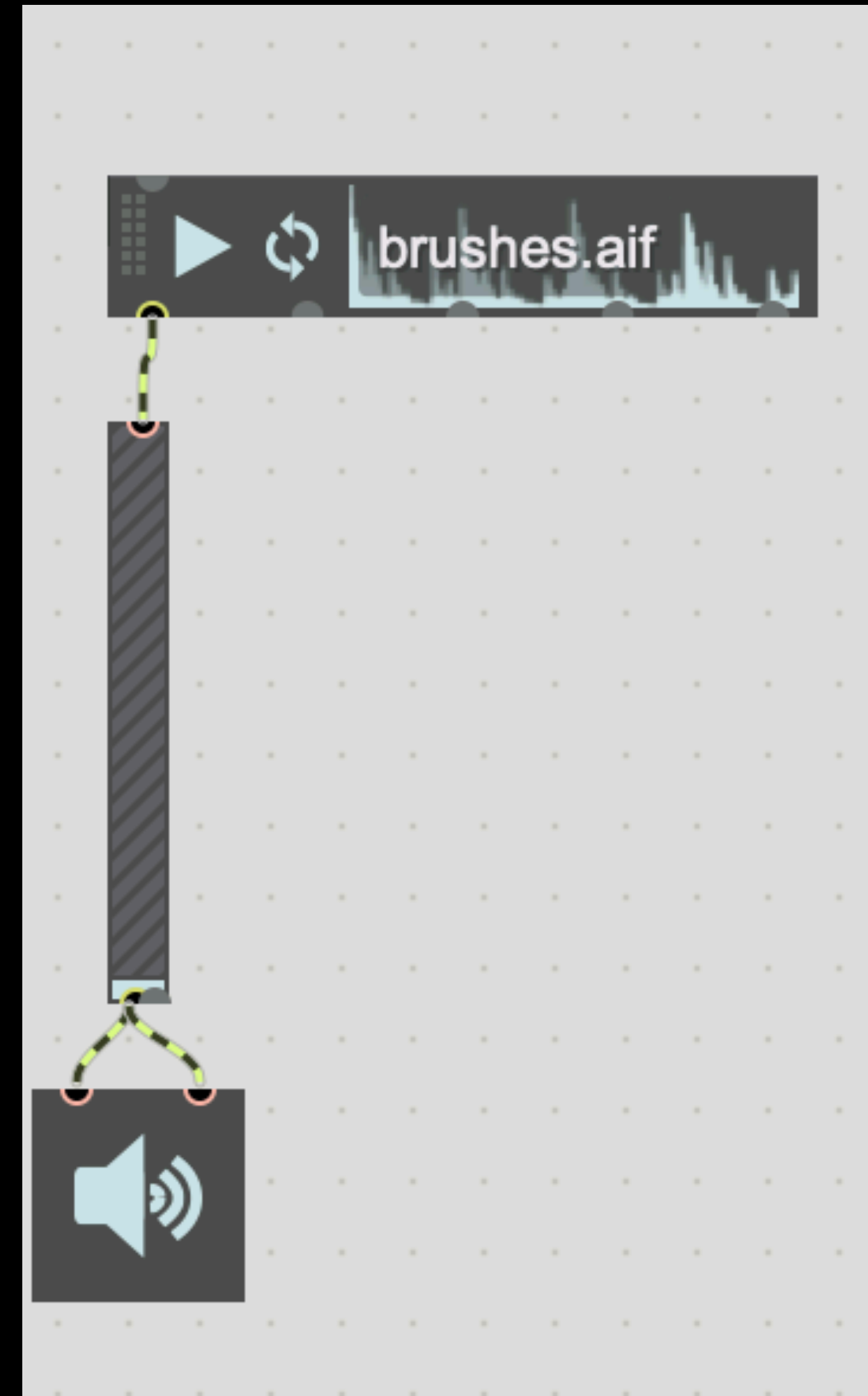
Drop a second sound on top of it.

This makes a **Playlist** object. Its one of many ways to play audio.



Add a **Gain** object to control volume. All DACs are linked, so its useful to give each its own volume.

Yellow patch cords mean its a signal.



Signal:

Yellow patch cords indicate a signal, aka audio.

Signals are steady streams of numbers, whereas normal data only happens periodically.

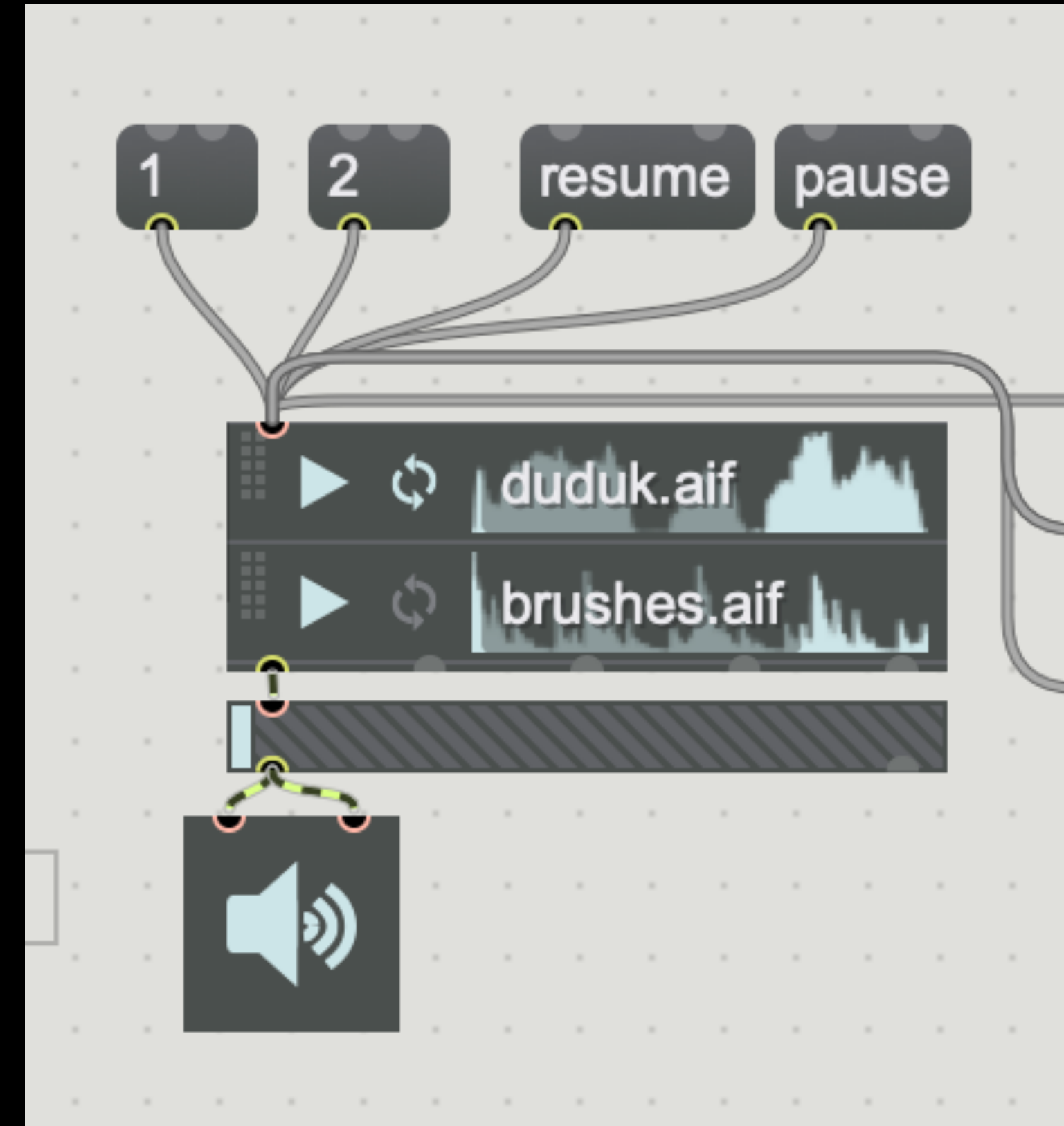
In order to produce a sound, a constant stream of numbers needs to be fed into the dac.



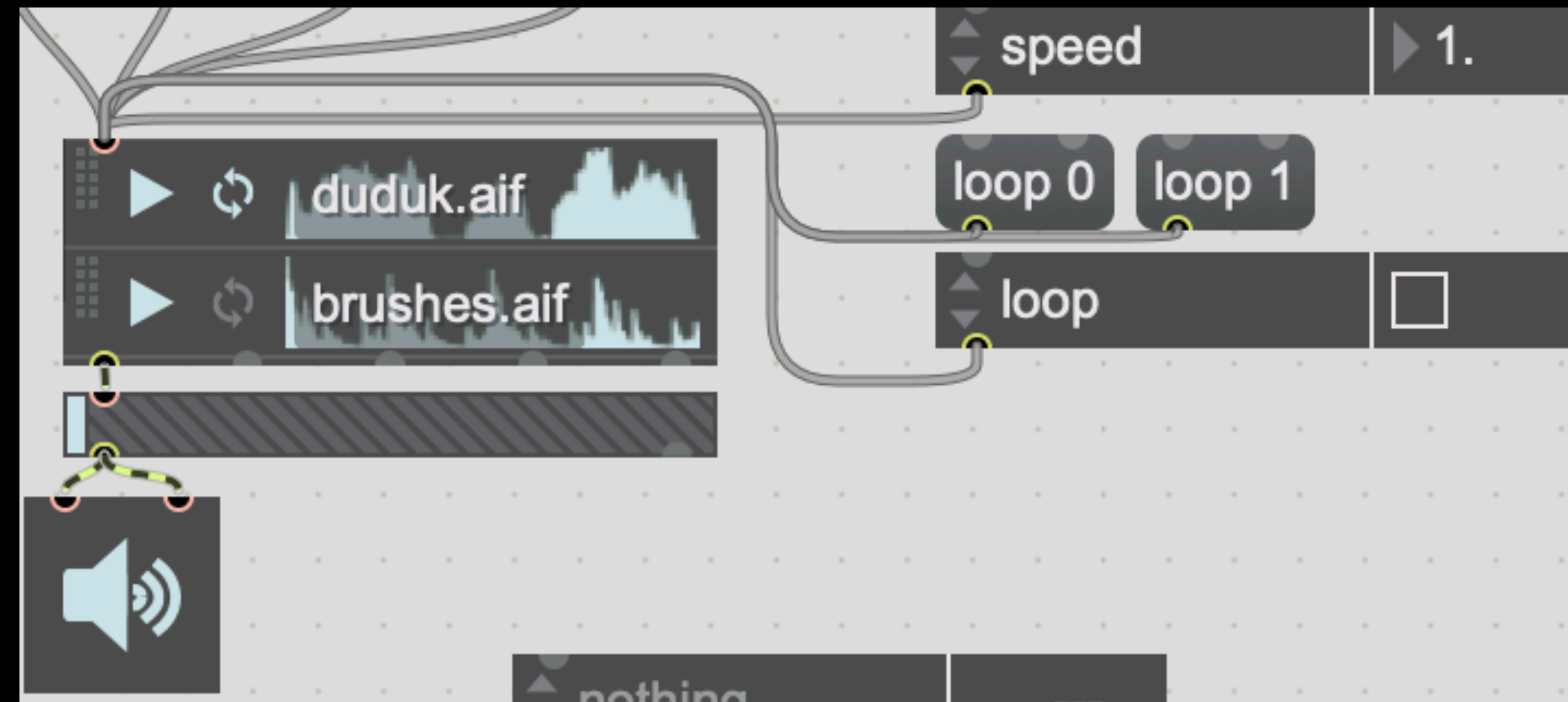
You can send messages to your play list to select which track to play using a number

You can pause/resume.

Check the help file for a full list.

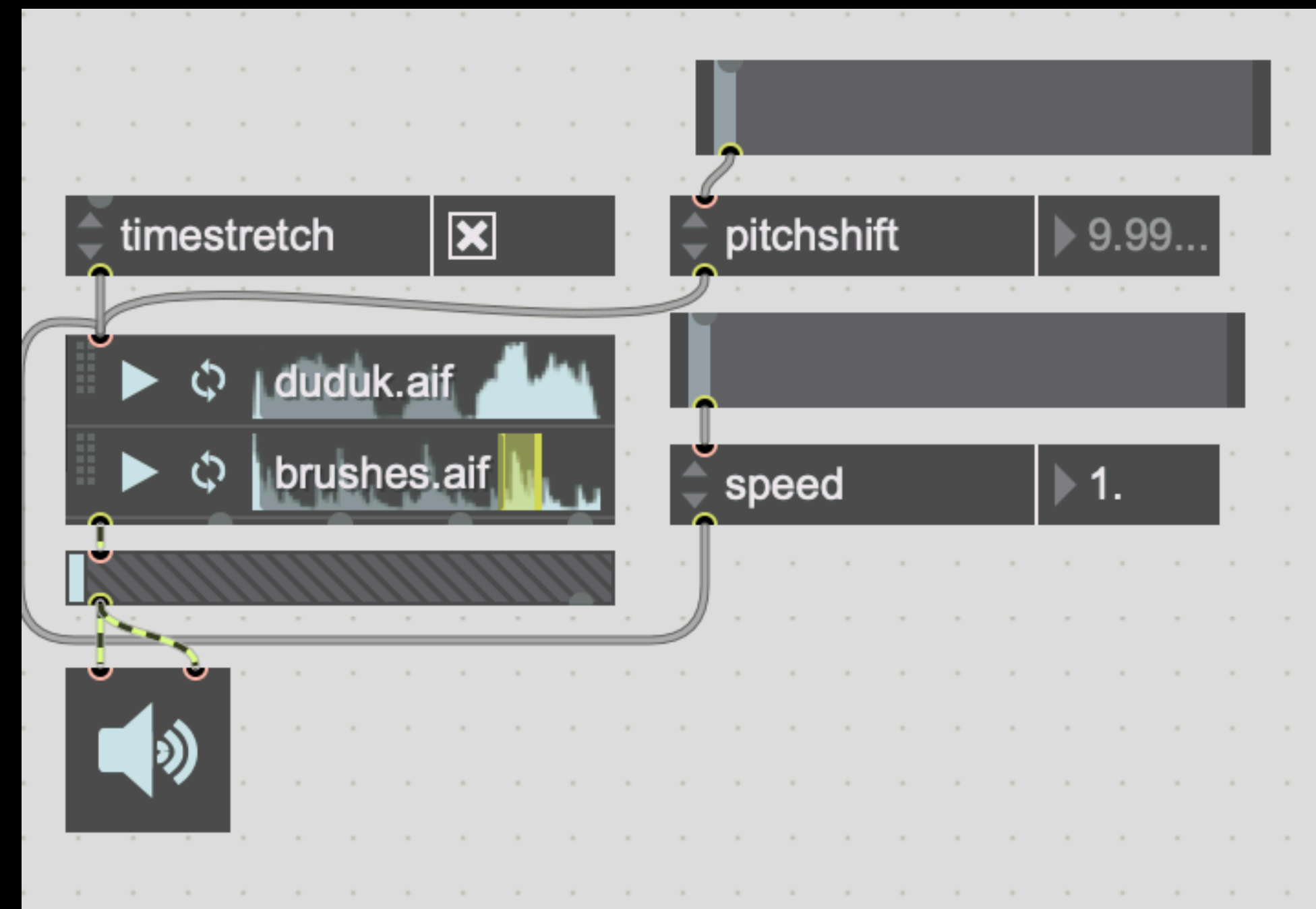


The attrui object (shortcut “a”) lets you change attributes. See help file for a full list

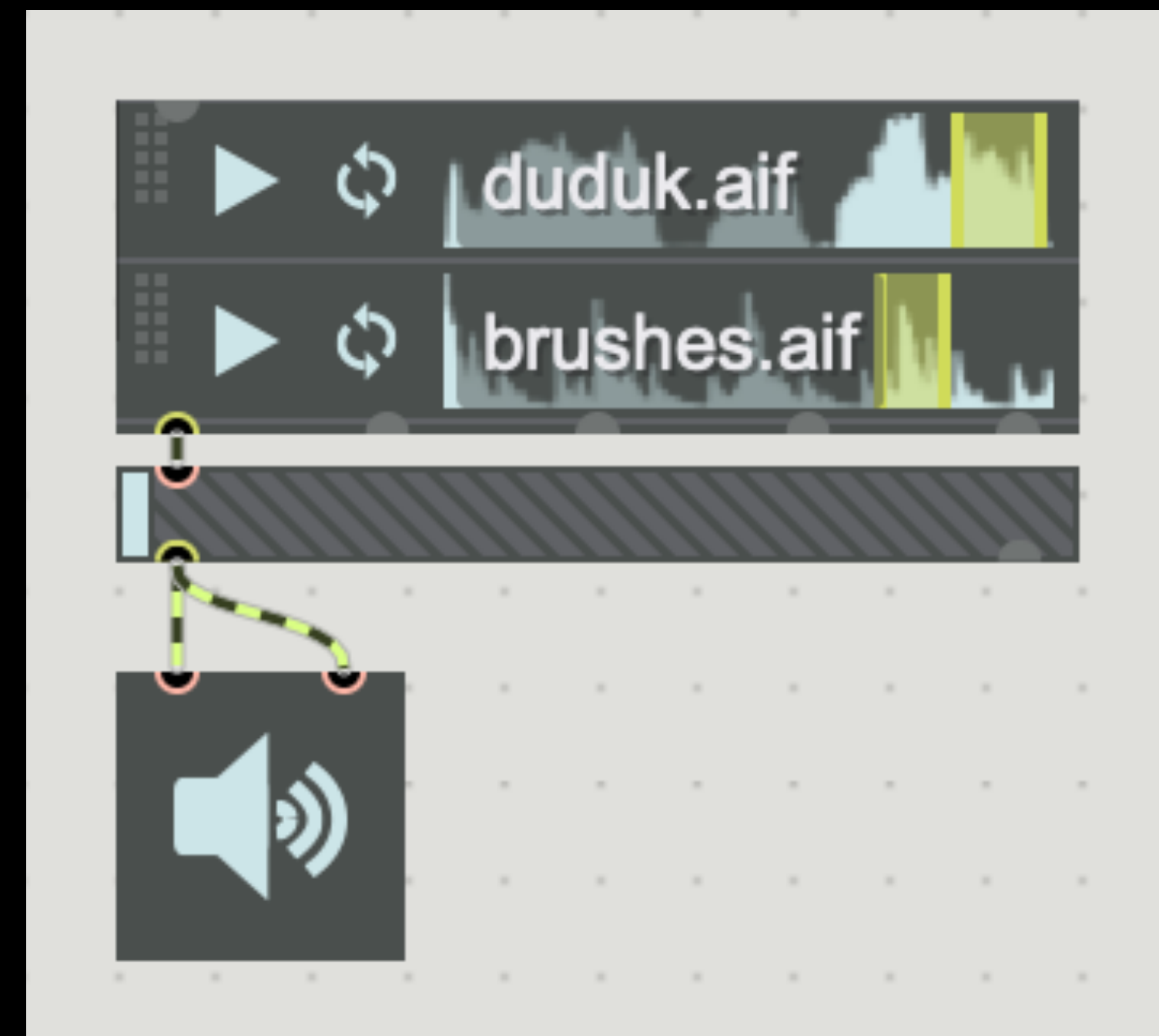


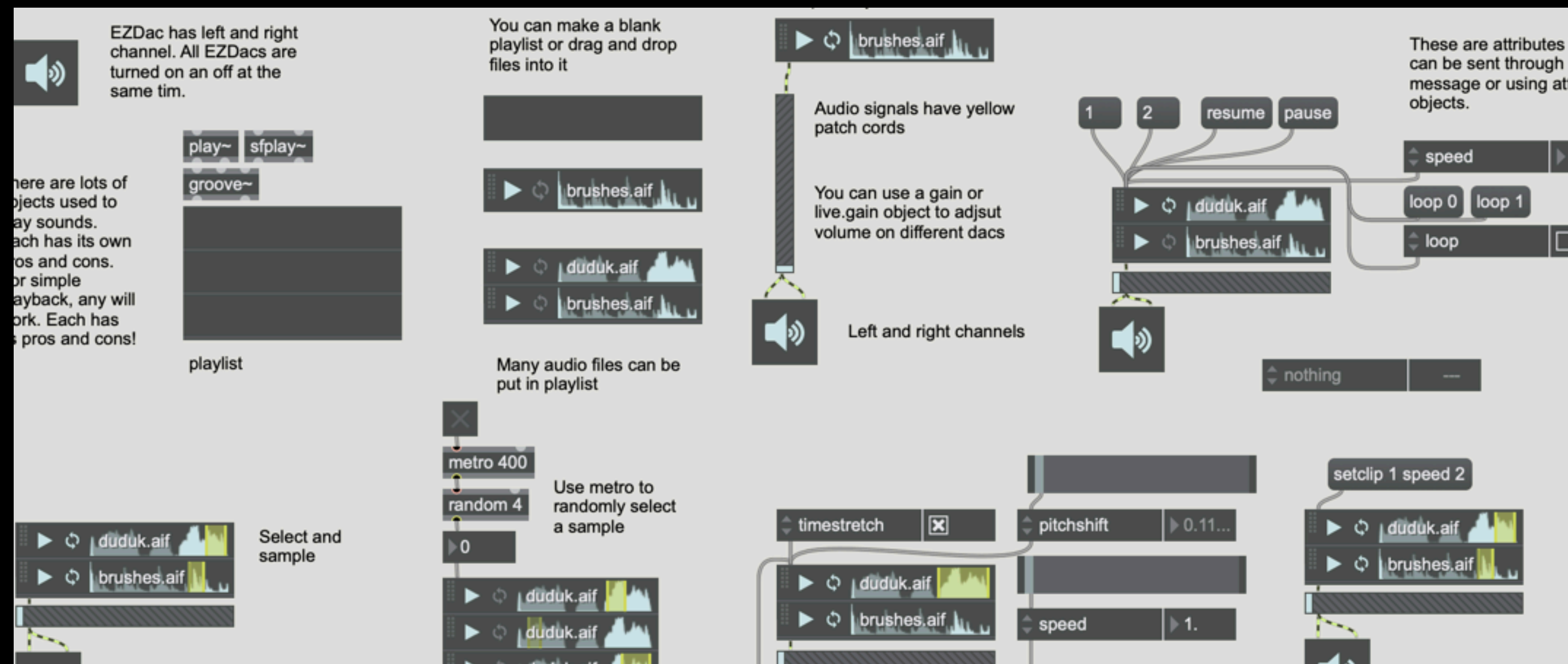
Enabling **timestretch** attribute on playlist lets you change the **pitchshift** and **speed**, all using attrui elements.

You can use a slider, don't forget to adjust the range!



Select sections for
sampling

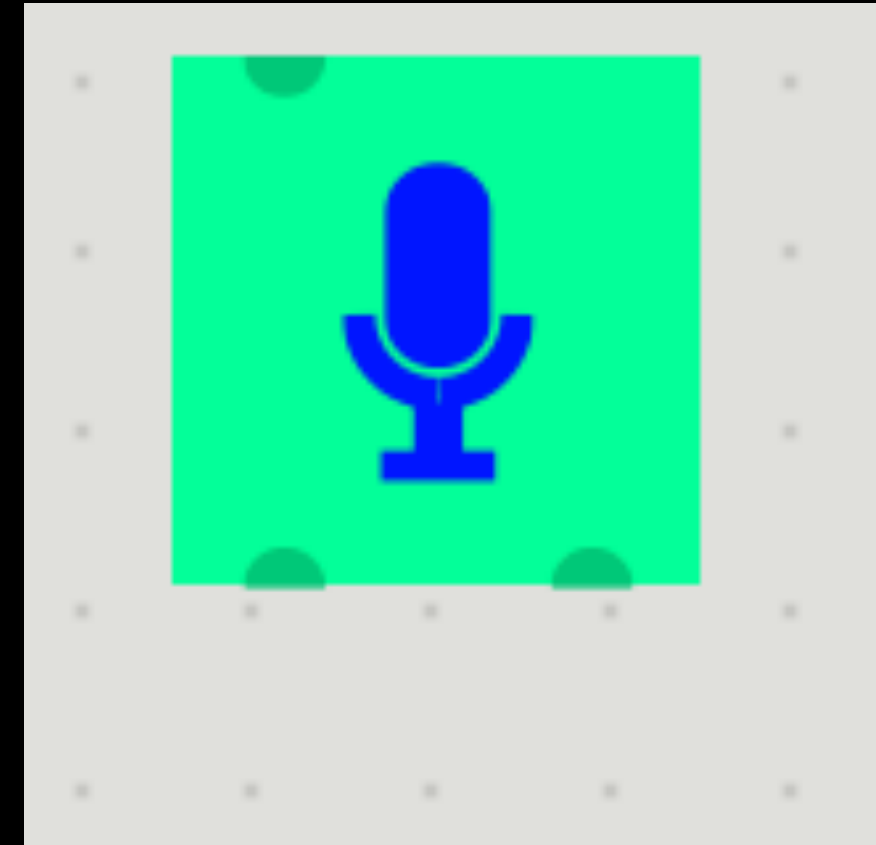




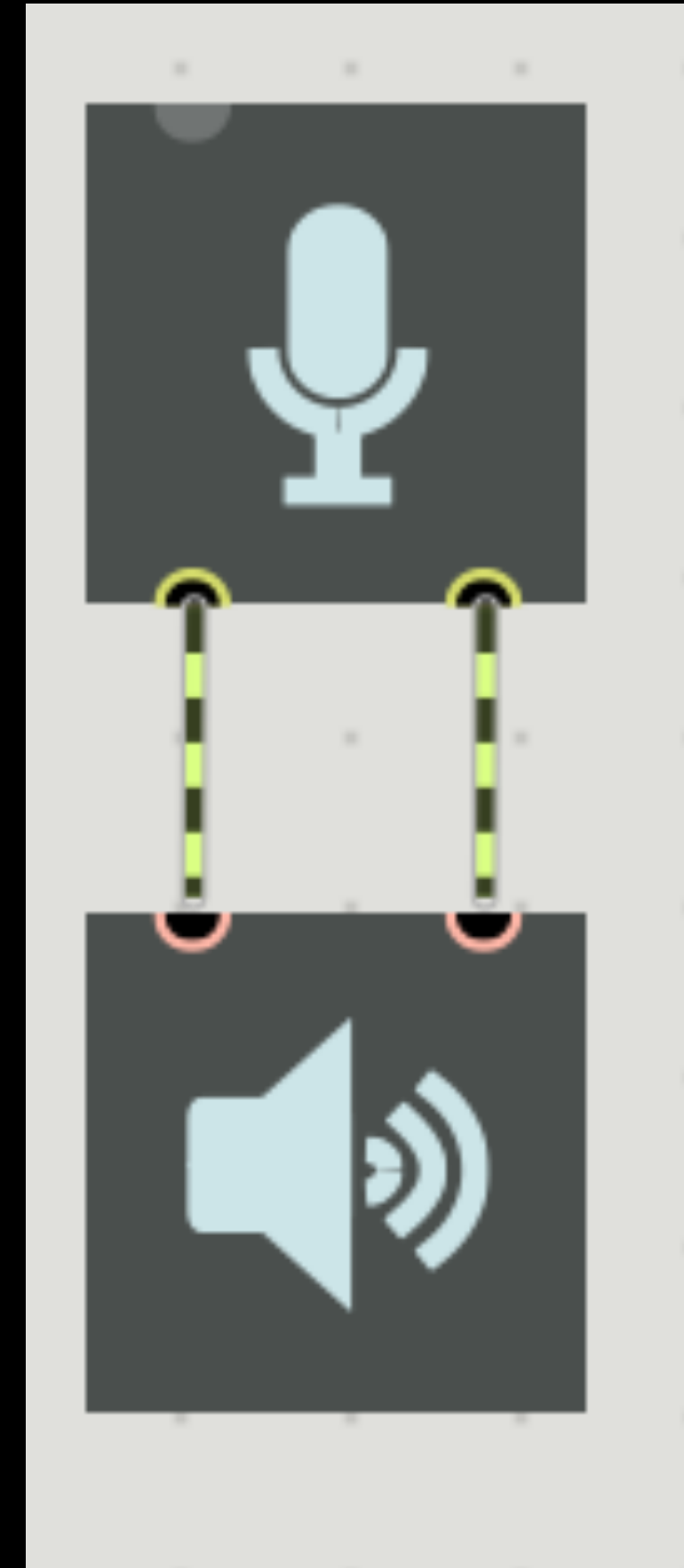
See all this in
using_playlists.maxpat

Recording Audio

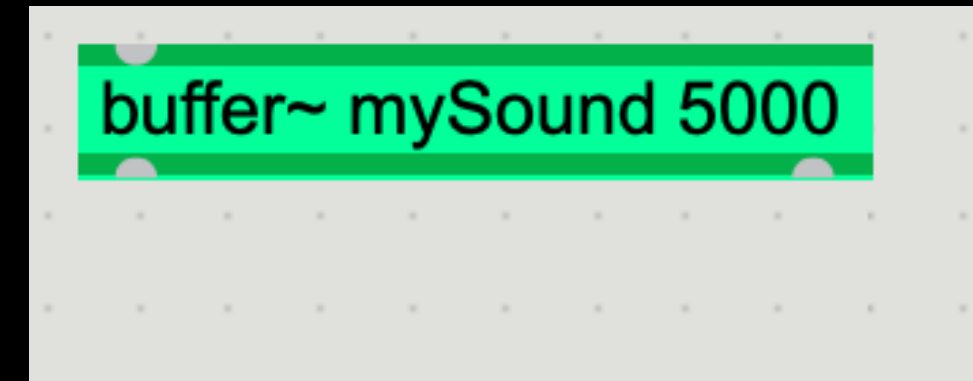
Use `ezadc~` to
have audio input.



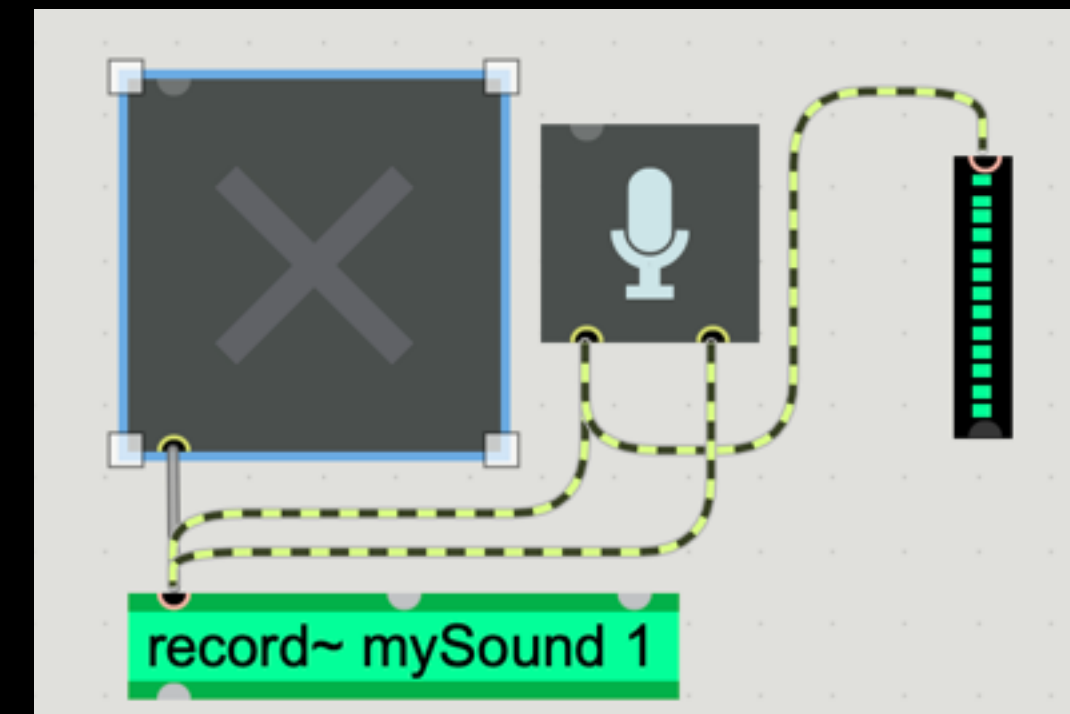
Connect ezdac~ to
ezadc~ to put the
audio recording
directly to the
speakers



Create a **buffer~** object to store your sound. Give it a size.

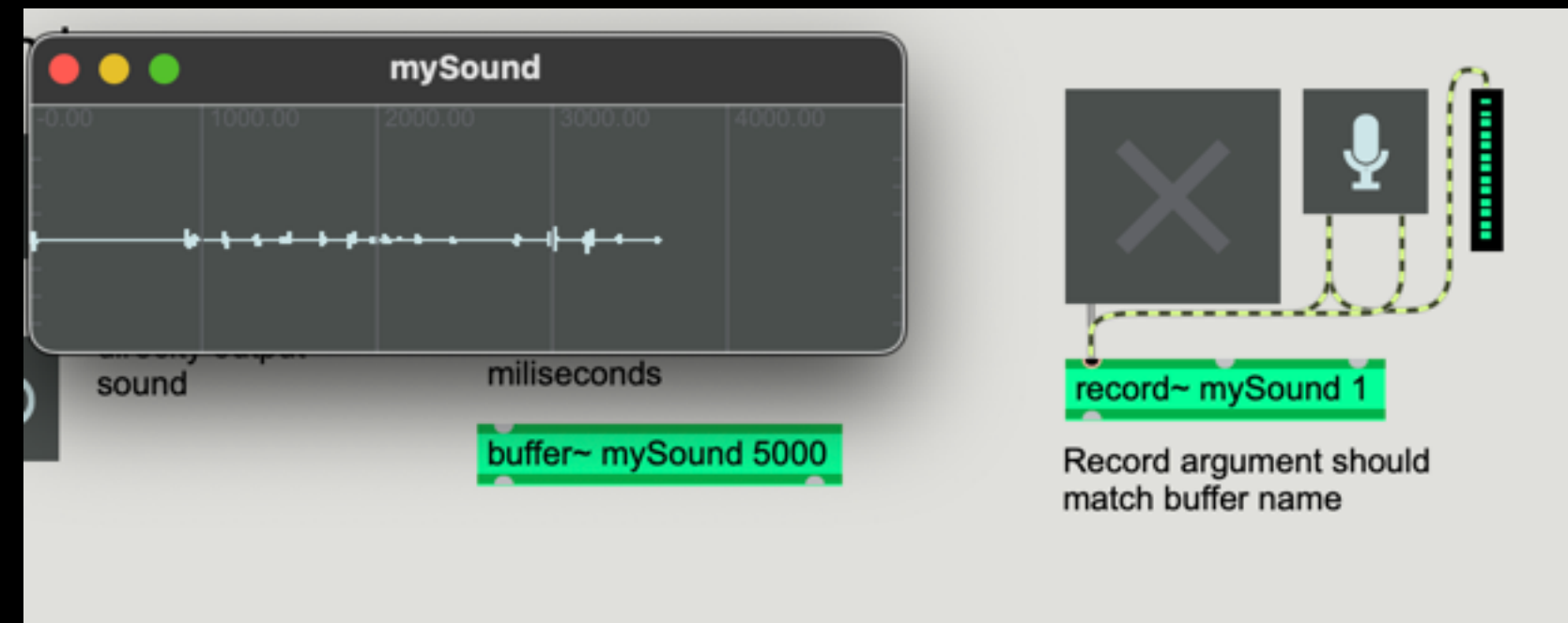


Then lets record. Use a **record~** object, connect a toggle and a **ezadc~**. Meter can help show you that sound is being input.

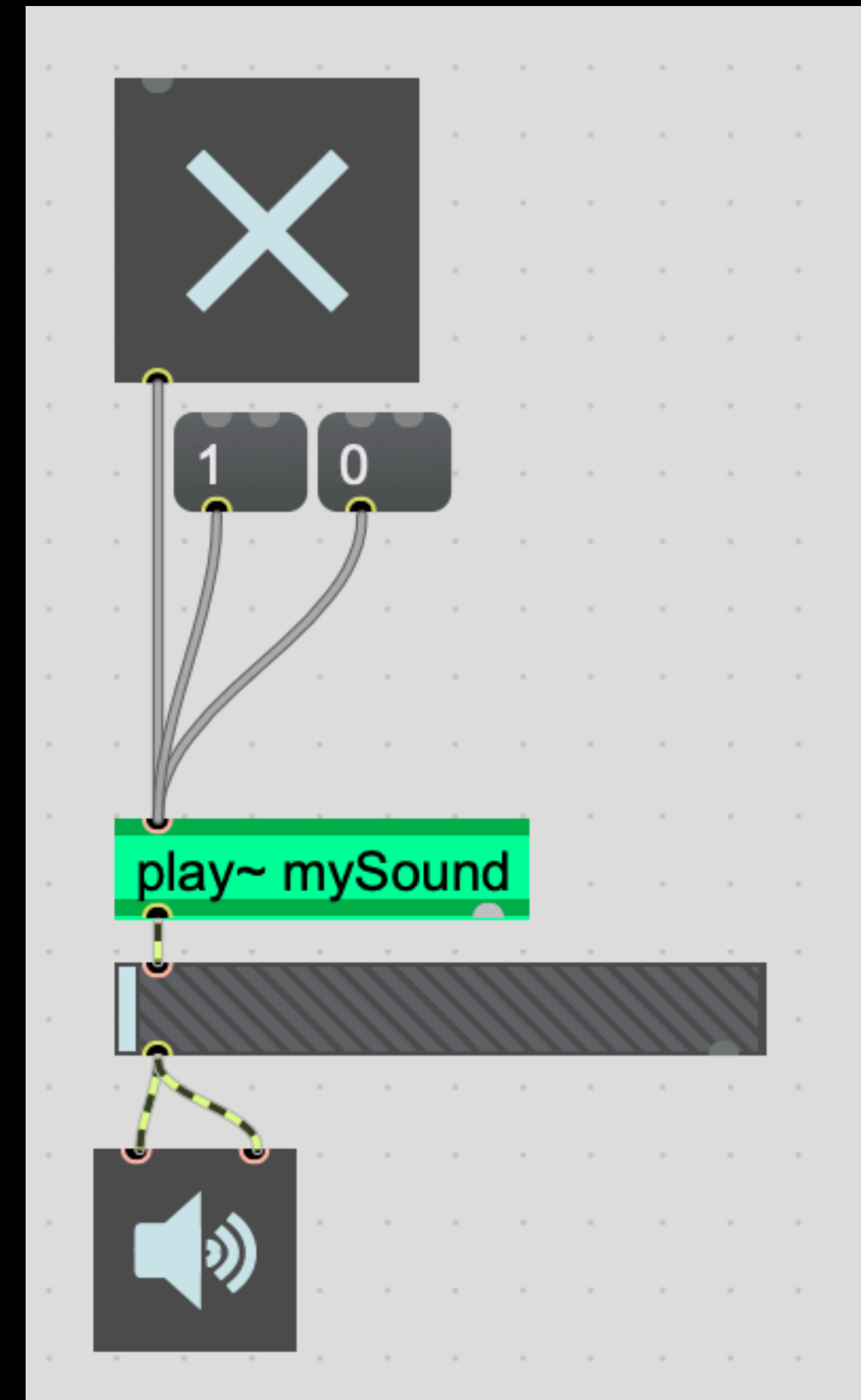


The buffer name must match the record name, you're referencing the same object!

Click the buffer
object to see the
waveform of your
recording

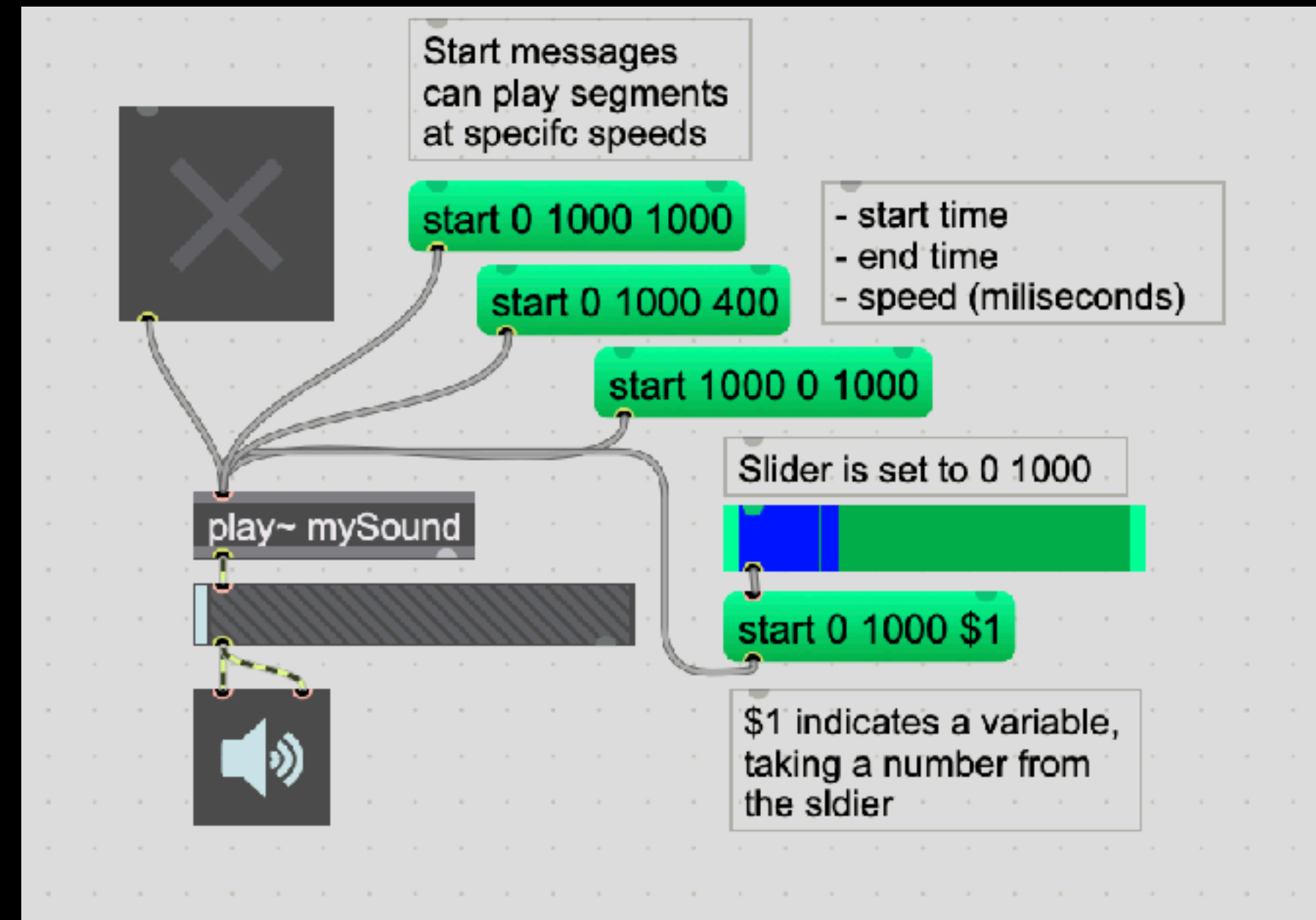


Play~ is an easy way to play this sound. Make sure it matches the buffer. Play needs a message of 0 or 1 to play. Toggle can do this, so can messages.

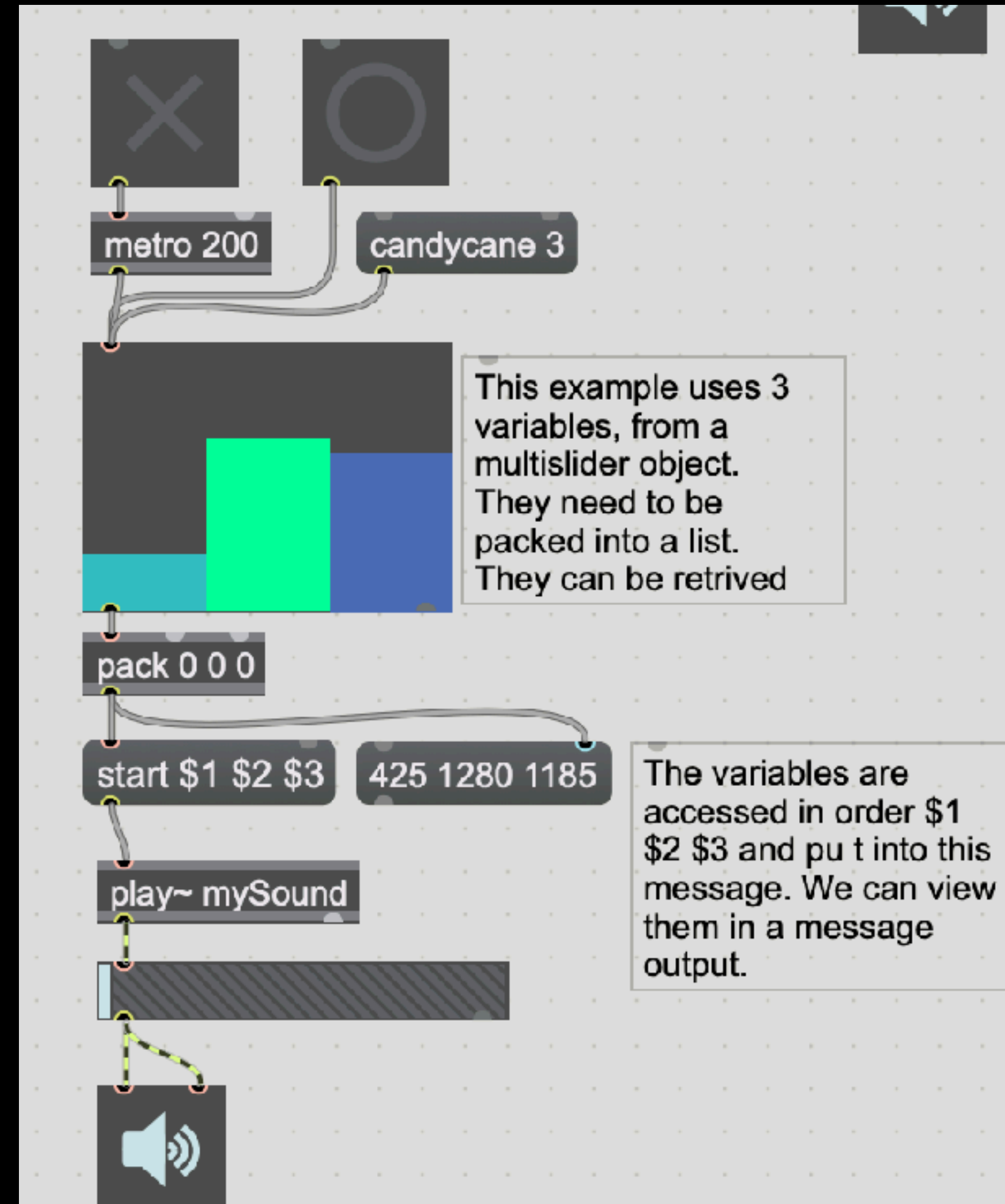


You can control play by setting a start, end, and duration in a message.

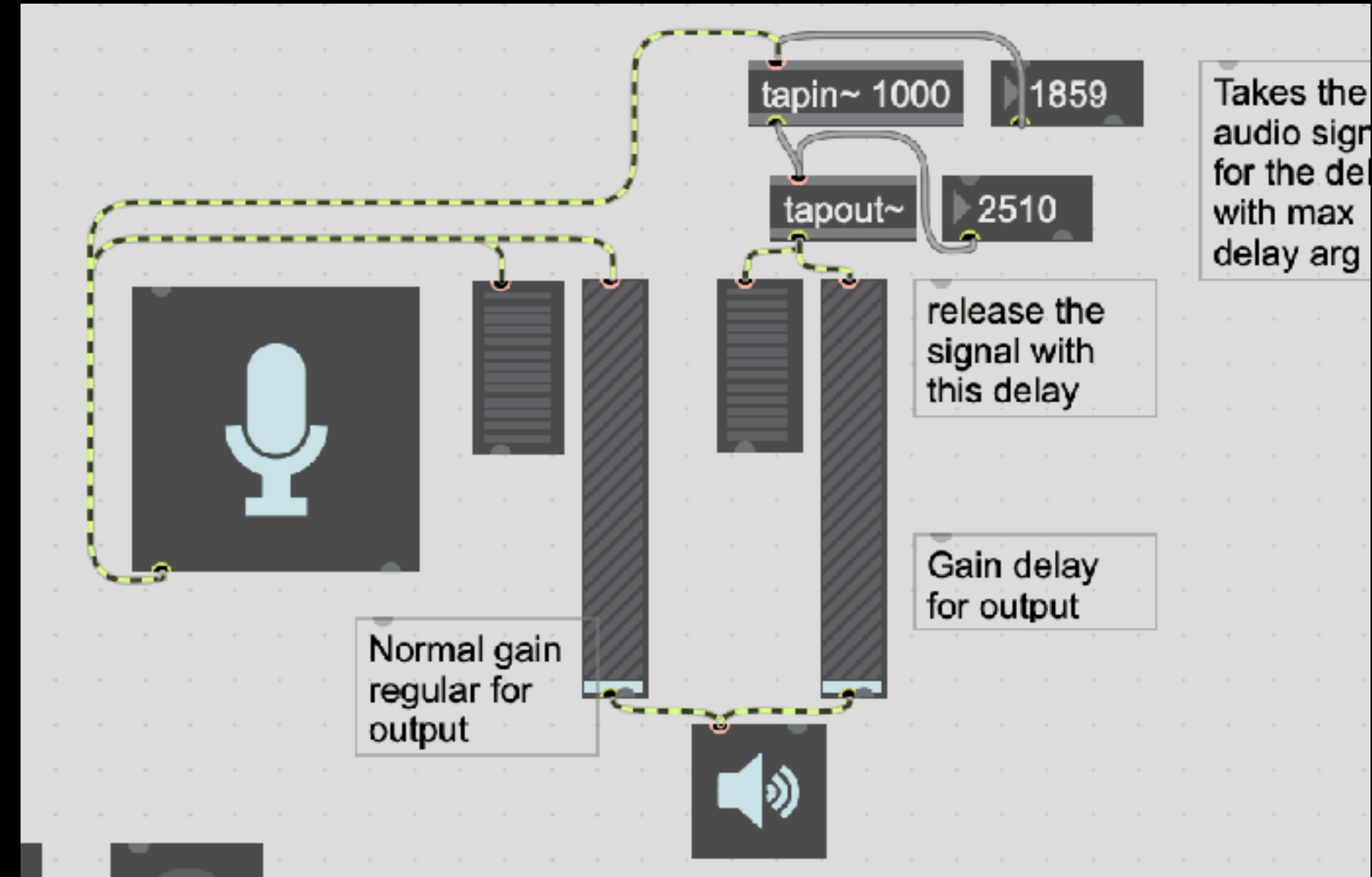
Here is an example with a variable and a slider. \$1 is a variable.

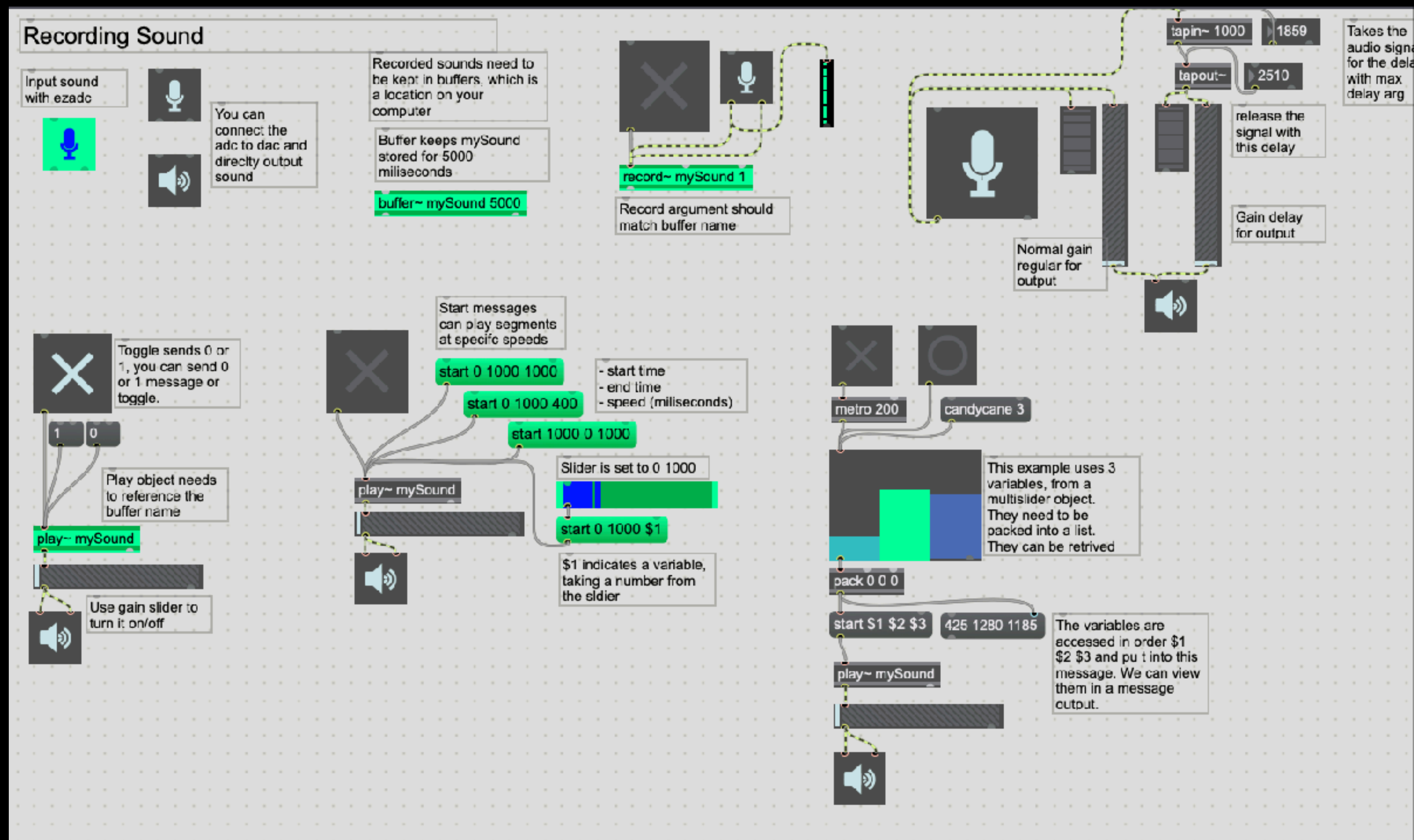


Click the buffer
object to see the
waveform of your
recording



Tapin/tapout can be used to create delay/reverb.



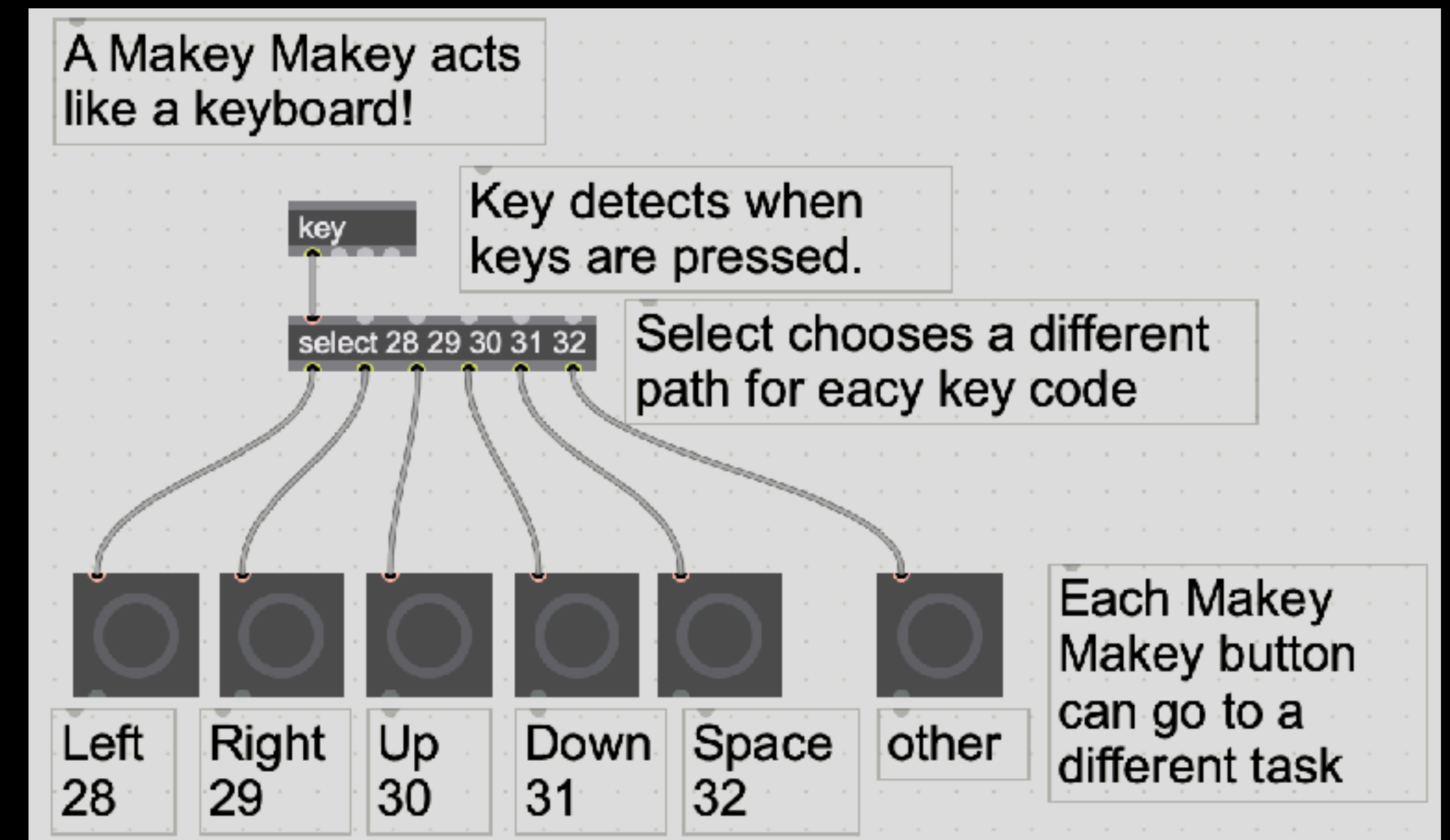


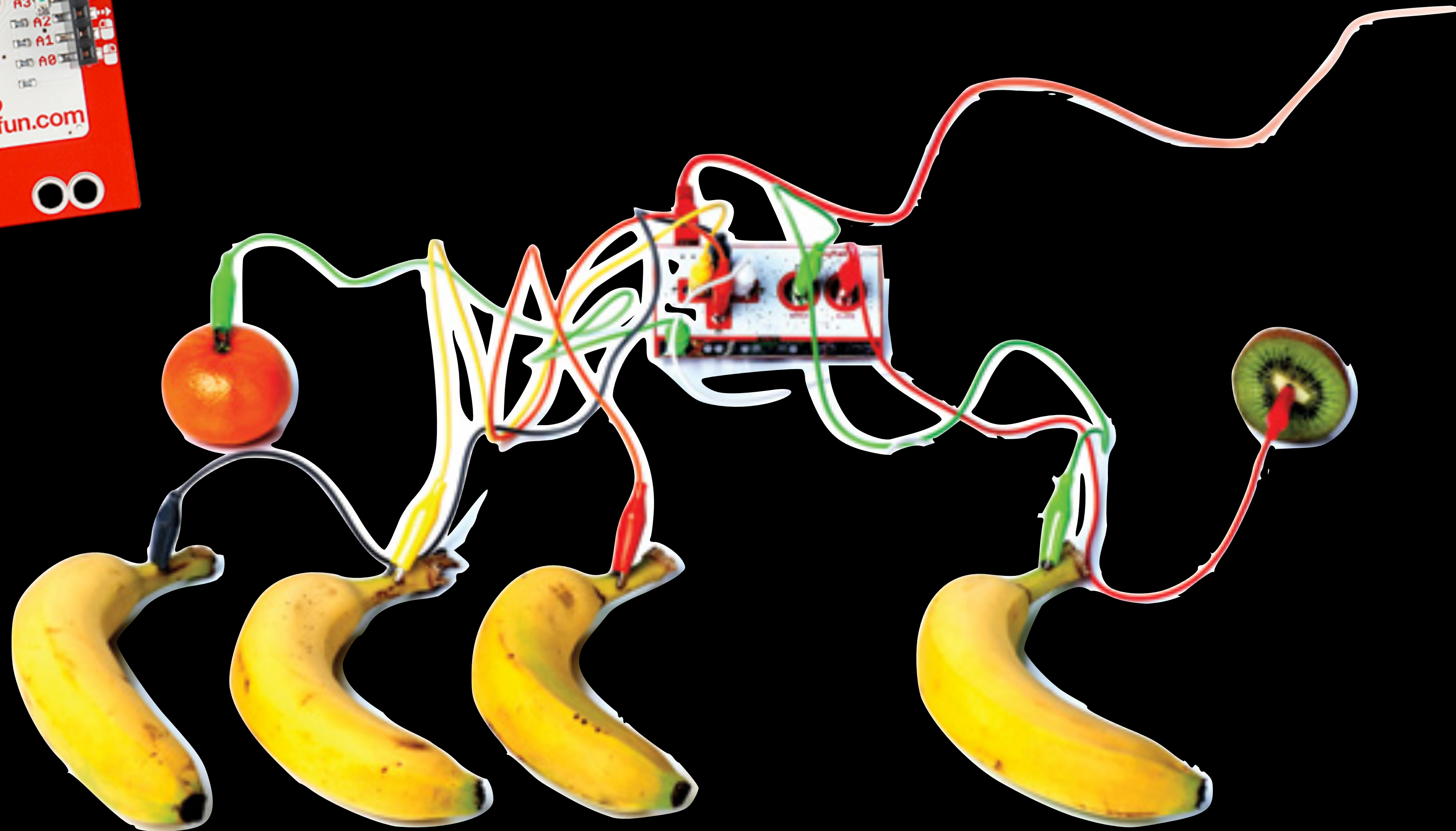
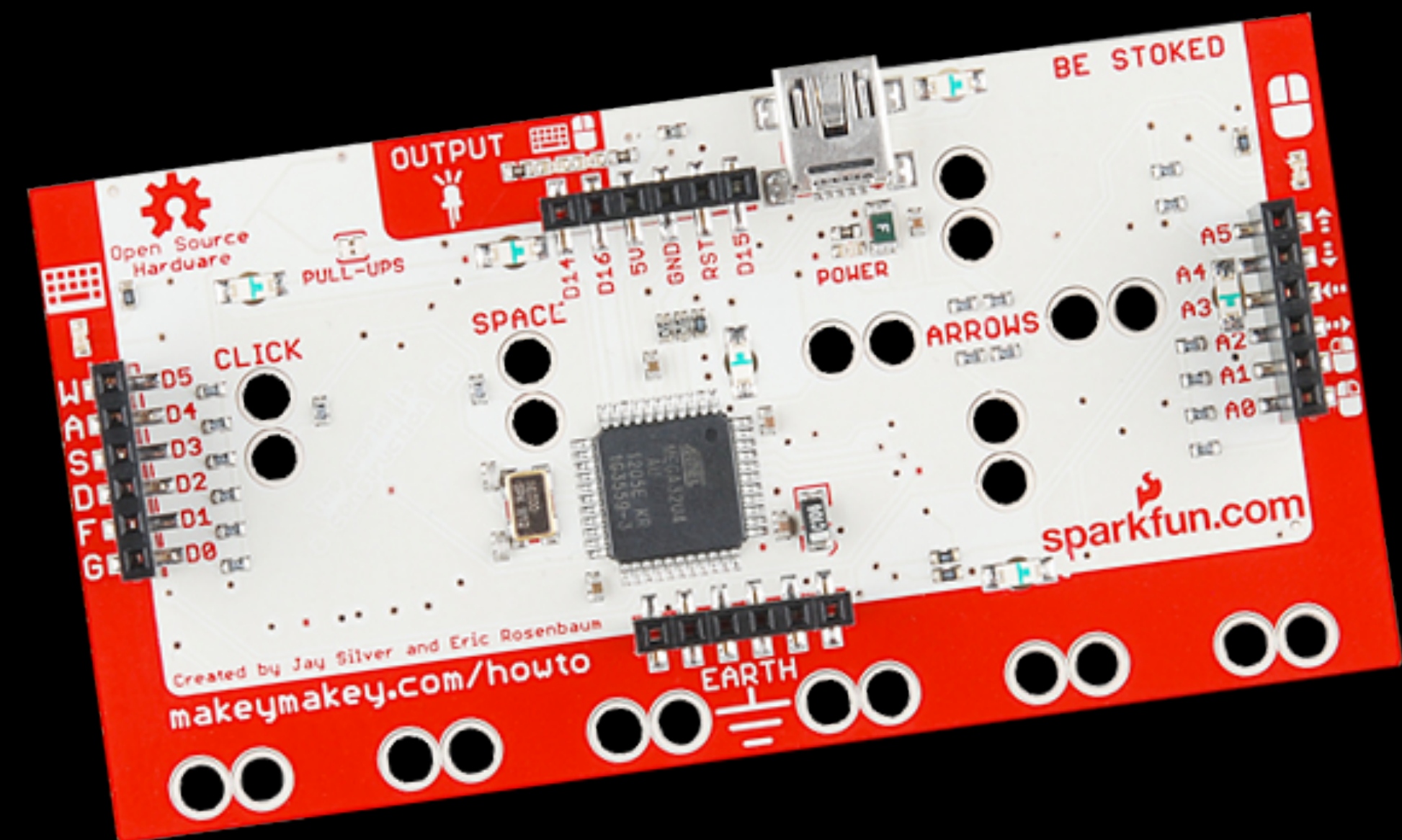
See all of this in
record_playback.maxpat

Introducing: Makey makey

This is a tool that simulates a keyboard using capacitive touch. You can essentially turn almost any object into a keyboard. It shows up as a keyboard on your computer.

Open `makey_makey.maxpat`





Makey Makey Exploration Time:

In groups, make something with the
MakeyMakey

Think of these as tools to build / mix /
expand to create instruments,
interactions, experiences.

How can we get out of our desks and
experience these sounds?

What movements or interactions do
you imagine paired with sounds?

Briefly look at:

Groove~

Waveform~

Next week: Studio time, sequencers,
waveforms.

**Download today's patch
on Moodle**

Homework:

Work on Audio Experiment (due next week)

Begin to form teams for mid term (max 3 people, 2 is ideal)